SOFTWARE REQUIREMENTS ENGINEERING — TASK 3

Smart Meter Fault Reporting System

Artifact: Stakeholder Requirements Gathering Report & Questionnaires

1. INTERVIEW QUESTIONS & EXPECTED ANSWERS (10 QUESTIONS)

Part A: Open-Ended Interview Questions

#	TYPE	INTERVIEW QUESTION	EXPECTED ANSWER / RESPONSE
1	Open	What are the primary operational challenges you currently encounter when reporting smart meter failures?	Delayed technician dispatch, lack of status visibility, and communication breakdowns with customer support.
2	Open	How would you describe your ideal workflow for submitting and resolving a fault ticket?	A simple digital platform where I can upload photo proof, receive an instant ticket ID, and track real-time repair progress.
3	Open	What specific information do you require in the final repair verification report?	Root cause analysis, technician identification, parts replaced, total resolution time, and formal customer sign-off.
4	Open	How does a delay in resolving a meter fault impact your daily operations or household?	It causes inaccurate estimated billing, potential power outages, and financial uncertainty regarding actual energy usage.
5	Open	What escalation procedure do you recommend for unresolved high-priority tickets?	Automatic escalation to a senior supervisor if no field technician is assigned within 24 hours of submission.

Part B: Closed-Ended Interview Questions

#	TYPE	INTERVIEW QUESTION	EXPECTED ANSWER / OPTIONS
6	Closed	On average, how many meter fault incidents do you handle or report monthly?	**1 – 5 faults** (Options: 1-5 / 6-10 / >10)
7	Closed	Do you currently possess access to a centralized dashboard for fault tracking?	**No** (Options: Yes / No)
8	Closed	What is the average resolution duration for a reported fault in the current system?	**More than 3 days** (Options: <1 day / 1-3 days / >3 days)
9	Closed	Which primary channel do you prefer for real-time status alerts?	**Mobile App Notifications** (Options: SMS / App / Email)
10	Closed	Does the existing system allow direct image uploads of the damaged meter screen?	**No** (Options: Yes / No)

2. WEB FORM QUESTIONS & EXPECTED ANSWERS (10 QUESTIONS)

Part A: Open-Ended Web Form Questions

#	TYPE	WEB FORM QUESTION PROMPT	SAMPLE USER INPUT / RESPONSE
11	Open	Please describe any specialized features you would like integrated into the digital web form.	GPS auto-location detection and automatic meter ID scanning via mobile camera.
12	Open	What is the most frequent technical error code or physical defect observed on your meter?	Screen freeze showing zero credit balance despite recent recharge.
13	Open	How can we streamline the form submission process to minimize filing effort?	Reduce compulsory text fields and utilize pre-populated dropdown categories for fault types.
14	Open	Provide feedback regarding field maintenance team responsiveness during weekend incidents.	Response times are excessively slow during weekends and official public holidays.
15	Open	What additional support resources should be accessible directly through the web form portal?	An automated AI Chatbot for immediate basic hardware troubleshooting steps.

Part B: Closed-Ended Web Form Questions

#	TYPE	WEB FORM QUESTION PROMPT	SELECTED ANSWER / OPTIONS
16	Closed	Rate the overall user-friendliness of the current web reporting interface:	**Poor** (Options: Excellent / Average / Poor)
17	Closed	Select the hardware classification of the affected smart meter:	**Single-Phase Meter** (Options: Single-Phase / Three-Phase)
18	Closed	Do you receive immediate automated confirmation via email/SMS upon submission?	**No** (Options: Yes / No)
19	Closed	Level of satisfaction with the current fault reporting and resolution lifecycle:	**Dissatisfied** (Options: Satisfied / Neutral / Dissatisfied)
20	Closed	Would you utilize a live-chat feature for instant technical guidance if integrated?	**Yes** (Options: Yes / No)

3. REQUIREMENTS GATHERING SYNTHESIS REPORT

Interview Opinions

- **Manual Inefficiency:** Universal consensus that paper/phone reporting is slow and prone to human error.
- **Need for Transparency:** Stakeholders strongly demand clear, traceable progress updates for submitted tickets.
- **Standardized Categorization:** Technicians emphasize the need for pre-defined fault codes to properly prioritize repairs.

Interview Feelings & Sentiments

- **Frustration:** High levels of user dissatisfaction due to long turnaround times and lack of response.
- **Anxiety:** Concerns regarding inaccurate estimated billing and unexpected power disconnections during fault periods.
- **Optimism:** High enthusiasm for an automated digital platform that guarantees accountability and tracking.

Current State of the System

- **Disjointed Channels:** Heavy reliance on manual phone calls, paper logs, and uncoordinated email threads.
- **Zero Real-Time Tracking:** Absence of digital status dashboards for consumers or supervisory staff.
- **Data Gaps:** Inability for management to extract performance metrics or track technician response efficiency.

System Goals (Future State)

- **End-to-End Automation:** Automated ticket creation, GPS routing, and instant assignment to nearby field technicians.
- **Real-Time Transparency:** Live status tracking (Pending → Assigned → In Progress → Resolved) for all users.
- **Performance Analytics:** Management dashboard for SLA monitoring, response tracking, and predictive maintenance.